Syllabus

COSC 3430, Artificial Intelligence, Fall 2026 | Augustana U.

Course Info

- Where: FSC 372 | When: MWF 9:00-9:50 am
- Canvas: augie.instructure.com/courses/15660
- AI use: permissive (details below)

Instructor

- Office/student hours: in room FSC 386 - book via Navigate
- Tue 1:00-3:00 pm, Thu 9:00 am-12:00 pm
- Phone: 605.274.4285 | Email: marcus.birkenkrahe@augie.edu
- More information: birkenkrahe.github.io or linkedin.com/in/birkenkrahe

Course Description

This course introduces the student to various aspects of artificial intelligence (AI), whose goals are the creation of more useful machines by making them more "intelligent." The course focuses on the fundamentals of machine learning (ML) and uses supervised, unsupervised, and reinforcement learning algorithms for classification and prediction tasks. The student will learn to build models from data using regression, logistic regression, classification and regression trees, random forests, ensemble and boosted methods, neural network techniques, and deep learning using convolutional neural networks (CNN). These algorithms are then applied to the areas of machine vision, image feature recognition, natural language processing (NLP), and general predictive techniques used in the field of Data Science.

Course AI Classification

AI Use in this Coursework: Permissive.

Students may use AI tools freely throughout this course, including on assessments, unless otherwise specified, provided use is disclosed and outputs are critically evaluated. AI tools are not permitted during exams unless explicitly stated on the exam itself.

Course Outcomes

Students who complete this course will be able to:

- Explain core machine learning paradigms (supervised, unsupervised, reinforcement learning)
- Implement regression and classification models
- Evaluate model performance using appropriate metrics
- Apply tree-based and ensemble methods
- Perform clustering and dimensionality reduction
- Build and train neural networks and CNNs
- Describe and implement basic reinforcement learning methods
- Analyze limitations, assumptions, and failure modes of AI systems

Prerequisites

COSC 2230 or COSC 1300

- **COSC 2230 (CS III):** You can write programs that don't just work, but scale, perform, and hold up under real-world conditions, using OOP design, generic functions, concurrency, core data structures and algorithms. Usually taught with Java.
- **COSC 1300 (Intro to DS):** Know tools to extract knowledge from data, manipulate, model and visualize data using Python and/or R.

University Policies and Resources

Use this link to access university-wide academic policies (Accessibility & Accommodations, Canvas, Honor Code, Illness Prevention & Protection, Student Well-being/Mental Health, Title IX Statement, and other Campus Resources including Help Desk, Registrar's Office, & Writing Center) are covered in a separate university-wide policy document.

Course Policies and Expectations

Optional policies are noted where applicable.

Attendance/Punctuality

This is a face-to-face course and you need to attend class to be successful in this topic. Many of our assignments will be started as in-class activities and will require your presence. If you are absent, make arrangements with a classmate to get class notes and a briefing on discussion topics, watch session recordings if available, and review the online material. Classes will start at the scheduled time and you should arrive 2-3 minutes prior to the start of class.

Absences

Let me know via email if you cannot attend a class. Augustana University only recognizes University-related absences and absences due to medical illness (you must provide a doctor's note).

Generative Artificial Intelligence Policy

AI tools predict, they don't think, and they will confidently hand you wrong answers alongside right ones. Learning to verify, correct, and take responsibility for AI-assisted work - propose, test, find where it breaks, revise - is part of what this course teaches, and it's also just good scientific practice. What isn't allowed is using AI as a substitute for that judgment: undisclosed use, or submitting output you haven't checked, defeats the purpose. For more details on "should you use AI to help you code", see: github.com/birkenkrahe/org/…/UsingAItoCode.pdf

Course Modification (inclement weather, illness, etc.) Policy

If significant inclement weather, professor illness, or any special circumstance arises that may require an adjustment to the course schedule (i.e. an alternative assignment, switching to synchronous online, canceling the class, etc.), every attempt will be made to communicate such changes to students by the night before the scheduled class. Communication to students will come via an announcement on the class LMS and/or an email to your AU email address. If you have reason to believe the class may not be able to be offered as scheduled, please keep an eye on these inboxes.

Exam Policy

Exams during the term are based on material discussed in class. Make-up exams will only be given if a written excuse (a physician's note or evidence of a University-excused absence) is submitted to me prior to the exam. Make-up exams may be different than in-class exams and the material may generally be more difficult. During exams, no cell phones, calculators, or other electronic devices may be used. All notes and books must be out of sight, except for materials allowed (as specified during the review session). You may not leave the room during the exam.

Course Materials and Access

Required textbooks

Aurélien Géron, "Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow" - notebooks and code at github.com/ageron/handson-mlp

Optional additional textbook

Simon J. D. Prince, "Understanding Deep Learning" - free and online with many resources at udlbook.github.io/udlbook

Course platforms

- GitHub for lectures, code: github.com/birkenkrahe/cosc_3430_fall_26
- Google Chat Space for discussion and sharing: tinyurl.com/gspace-3430
- Google Drive for whiteboard photos and papers: tinyurl.com/fa26-photos-3430
- DataCamp for additional optional practice: datacamp.com
- YouTube Playlist: tinyurl.com/youtube-3430

Assignments and Grading Procedures

Grading table

Letter Grade	Percentage Range
A	93-100%
A-	90-92%
B+	87-89%
B	83-86%
B-	80-82%
C+	77-79%
C	73-76%
C-	70-72%
D+	67-69%
D	63-66%

Letter Grade	Percentage Range
D-	60-62%
F	Below 60%

Grade breakdown

What	When	How much
Project (Scrum)	monthly	25%
Quizzes (not timed)	weekly	15%
Programming assignments	weekly	40%
Midterm exam (50 min)	TBD	10%
Final exam (optional)	TBD	10%

- **Programming assignments:** at home, individual, AI allowed unless specified - use must be disclosed, outputs critically evaluated and tested. Estimated time: 40-55 hrs/term (3-4 hrs/week)
- **Term Project** (team of 2-3): 3 sprint reviews (3 x 5%) + presentation (10%)
- **Quizzes:** multiple choice, not timed, at home, individual
- **Midterm exam:** 50 min, individual, based on quizzes (with mock exam)
- **Final exam:** 30 min, optional, oral

Course Schedule

One quiz and one programming assignment per week (see Grading). This schedule may be subject to changes depending on class progress.

Date	Topic	Notes
Wed (8/26)	Overview	Review the AI policy, clarify questions
Fri (8/28)	Setup/projects	
Mon (8/31)	Ch 1. ML fundamentals 1/3	
Wed (9/2)	Ch 1. ML fundamentals 2/3	Last day for online reg./adding w/o signature
Fri (9/4)	Ch 1. ML fundamentals 3/3	
Mon (9/7)	NO CLASS - Labor Day	
Wed (9/9)	Ch 2. End-to-end project 1/3	
Fri (9/11)	Ch 2. End-to-end project 2/3	Independent Study/Internship paperwork due
Mon (9/14)	Ch 2. End-to-end project 3/3	
Wed (9/16)	Ch 2. End-to-end project (cont.)	Last day to drop without signatures
Fri (9/18)	Ch 2. End-to-end project (cont.)	Extended chapel service (until 11:15a)
Mon (9/21)	Ch 3. Classification 1/3	

Date	Topic	Notes
Wed (9/23)	Ch 3. Classification 2/3	
Fri (9/25)	Ch 3. Classification 3/3	Sprint 1: Project Proposal
Mon (9/28)	Ch 4. Training models 1/3	
Wed (9/30)	Ch 4. Training models 2/3	Graduation Application Deadline (10/1)
Fri (10/2)	Ch 4. Training models 3/3	
Mon (10/5)	Ch 5. Decision trees 1/2	
Wed (10/7)	Ch 5. Decision trees 2/2	
Fri (10/9)	MIDTERM	
Mon (10/12)	NO CLASS - Native American Day	
Wed (10/14)	NO CLASS - Fall Break	
Fri (10/16)	Ch 6. Random forests 1/2	
Mon (10/19)	Ch 6. Random forests 2/2	
Wed (10/21)	Ch 7. Dimensionality reduction 1/2	
Fri (10/23)	Ch 7. Dimensionality reduction 2/2	Sprint 2: Literature Review / Method
Mon (10/26)	Ch 8. Unsupervised learning 1/3	
Wed (10/28)	Ch 8. Unsupervised learning 2/3	Last day to drop with "W" or file S/U
Fri (10/30)	Ch 8. Unsupervised learning 3/3	
Mon (11/2)	Ch 9. Shallow neural networks 1/2	
Wed (11/4)	Ch 9. Shallow neural networks 2/2	
Fri (11/6)	Ch 9. Deep neural networks 1/2	
Mon (11/9)	Ch 9. Deep neural networks 2/2	
Wed (11/11)	NO CLASS - Veterans Day	
Fri (11/13)	Ch 10. Intro to PyTorch 1/2	
Mon (11/16)	Ch 10. Intro to PyTorch 2/2	Interim 2027 & Spring 2027 registration (11/16-20)
Wed (11/18)	Ch 11. Training Neural Networks 1/3	
Fri (11/20)	Ch 11. Training Neural Networks 2/3	Sprint 3: Results / Abstract
Mon (11/23)	Ch 11. Training Neural Networks 3/3	
Wed (11/25)	NO CLASS - Thanksgiving Break	
Fri (11/27)	NO CLASS - Thanksgiving Break	
Mon (11/30)	Presentations	Sprint 4: Discussion
Wed (12/2)	Presentations	

Date	Topic	Notes
Fri (12/4)	Presentations	Last day of classes
(12/7-12/11)	FINAL EXAM	Date/time/location TBD
		Final grades due Wed (12/16)

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